

Beyond the Raw Waveform: Fusing Visual Representations of EDA for Stress Detection

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Abstract—Electrodermal activity (EDA) is widely used in automatic stress detection, yet most pipelines treat it only as a raw one-dimensional waveform. This study examines whether complementary visual representations of EDA provide useful information for stress classification and whether their fusion improves recognition performance. Six image-based representations are derived from each EDA recording: an unwrapped short-time Fourier transform (STFT) phase spectrogram, an instantaneous-frequency map computed from that phase, a power spectral density (PSD) spectrogram, a continuous wavelet transform scalogram, a recurrence plot, and a rendered waveform trace. The selected representations are stacked as channels of a single multichannel input, together with the raw waveform, and processed by a shared asymmetric-attention architecture. Experiments on a 58-subject stress dataset show that representation fusion improves over the raw waveform. The best configuration, which combines five representations while excluding the unwrapped phase spectrogram, reaches 70.97% test accuracy, compared with 67.36% for the raw waveform. The single PSD spectrogram achieves 69.44%, remaining close to the best-fused configuration at a lower computational cost. The results show that alternative visual forms of the same EDA signal can provide useful inductive biases for stress detection, and that a compact selection of complementary representations can be more effective than the raw waveform alone.

Index Terms—Stress recognition, electrodermal activity, signal-to-image representation, multi-representation fusion, transformer

I. INTRODUCTION

Stress is a coordinated physiological and psychological response to perceived demands. It engages the autonomic nervous system and activates neuroendocrine pathways whose intensity and duration depend on the stressor and the individual response [1], [2]. These responses may appear as brief situational episodes or develop into prolonged chronic states, with different physiological patterns and health consequences. Questionnaire instruments such as the Perceived Stress Scale are widely used to quantify subjective stress in clinical and research settings [3]. However, retrospective self-report is vulnerable to recall bias and cannot capture fine temporal changes in stress levels [4]. Salivary cortisol provides a validated neuroendocrine marker of stress, but sample collection is intrusive, and cortisol dynamics are too delayed for continuous real-time monitoring [2].

Stress has become a major public health problem. A large-scale analysis of nationally representative survey data from 146 countries reported that stress levels roughly doubled over an 18-year period, with widening demographic and socioeconomic disparities [5]. In occupational settings, psychosocial work-related exposures account for a measurable share of cardiovascular disease and depression cases across Europe [6]. Chronic psychological stress has also been associated with immune dysregulation, elevated cardiovascular risk, and

depressive disorder, reinforcing the need for reliable monitoring methods beyond occasional subjective assessment [7].

Continuous stress monitoring is especially important in settings where conventional assessment is difficult. In clinical, occupational, and operational environments, self-report can be delayed, incomplete, or affected by social desirability and demand characteristics. Wearable devices offer a practical physiological alternative, but user compliance, motion artifacts, and deployment variability remain persistent challenges [8]. Field studies have repeatedly identified data quality and signal integrity as key barriers to generalization outside laboratory protocols [9]. A useful stress-detection pipeline should therefore extract as much information as possible from the signals that wearable devices already collect, under conditions that approximate ambulatory use [10].

Electrodermal activity (EDA), also known as galvanic skin response, is among the most established peripheral signals for automatic stress monitoring. It reflects the activity of eccrine sweat glands, which are controlled by the sympathetic branch of the autonomic nervous system and are not under direct voluntary control [11], [12]. Functional neuroimaging studies have linked EDA to cortical and subcortical regions including the amygdala, insula, and anterior cingulate cortex, placing the signal within the broader neural circuitry involved in stress processing [13]. Most EDA-based systems still process the signal as a raw or minimally filtered one-dimensional waveform. Time-frequency and nonlinear-dynamics representations remain less explored, although they may expose structure that is difficult to learn from the waveform alone [14], [15]. Recent work in affective computing has also emphasized the importance of how behavioral and physiological information is organized before classification. Examples include graph-based facial representations for stress recognition, modality-agnostic physiological fusion for cognitive workload assessment, and lightweight modeling of pain-related brain activity [16]–[18].

This study examines representation choice and fusion for EDA-based stress detection. Six image-based representations are derived from each EDA recording, covering time-frequency and nonlinear-dynamics views of the same signal. These representations are evaluated individually and in channel-stacked combinations with the raw waveform, using a shared asymmetric-attention architecture without separate encoder branches or late fusion. The aim is to assess what is gained by moving beyond the raw waveform, and whether several complementary representations can improve stress recognition when treated as channels of a single multichannel input.

II. RELATED WORK

Electrodermal activity is widely used in affective computing because it is sensitive to sympathetic arousal and can be recorded continuously with wearable sensors [19]. In automatic emotion recognition, EDA is commonly used to estimate arousal and valence from tonic and phasic features of the signal. A recent systematic review and meta-analysis reported that EDA-based models perform more reliably for arousal than for valence, in line with the signal’s physiological

role as an autonomic arousal marker rather than a direct index of affective tone [20]. EDA is also used in automatic pain assessment, a related affective-computing task, where recent models have fused EDA with functional near-infrared spectroscopy through gradient-infused attention [21] and with electrocardiogram signals through cross-modal transformer fusion [22]. In these cases, EDA is usually treated as one physiological channel among several, rather than as a signal whose internal representations are studied in detail.

Pain and stress can elicit overlapping affective and autonomic responses. Accordingly, automatic pain assessment has been studied using a wide range of behavioral and physiological signals [23]. Electrocardiography (ECG) has been combined with demographic information in both single-task and multi-task formulations [24], [25], while facial video has been fused with heart-rate signals [26]. Other work has combined facial video with Functional near-infrared spectroscopy (fNIRS) [27], and a unified tokenization framework has recently been used to process facial videos, raw fNIRS waveforms, and fNIRS spectrograms within the same architecture [28]. fNIRS-specific models have also explored haemoglobin-difference representations for pain recognition [29].

Stress detection is one of the main application areas for EDA [14]. Recent work has increasingly applied deep learning directly to wearable-device EDA data, using either handcrafted descriptors or the complete raw signal in custom architectures for early stress detection. In some cases, EDA alone has outperformed multimodal combinations with electrocardiogram and photoplethysmography signals [15]. A smaller group of studies has converted EDA into image-based representations. Short-time Fourier transform and Mel spectrograms of the phasic EDA component, combined with convolutional networks such as VGG16, have improved emotion-classification accuracy over feature-based baselines [30]. Signal-to-image encodings have also been explored for stress detection, with Gramian Angular Field, Markov Transition Field, and recurrence plot representations compared on the WESAD dataset; Gramian Angular Field was the strongest encoding, although the analysis used multivariate physiological windows rather than EDA alone [31]. Outside the EDA and stress literature, fusing several signal-to-image transformations by stacking them as channels has been shown to outperform individual transformations and to extend naturally to multivariate and multi-sensor time-series classification [32].

A related strategy was explored in [33], where multiple EDA representations were generated and composited into a single multi-representation diagram for automatic pain recognition. This approach was competitive with and, in several cases, superior to conventional multimodal fusion. Image-domain physiological processing has also been used at larger scale: a vision foundation model trained across behavioral and physiological modalities, including EDA, electrocardiogram, electromyogram, and functional near-infrared spectroscopy, maps each modality into a shared image-based representation space for pain assessment [34]; similarly, a lightweight embedding model for biosignal analysis has been pretrained on 4.4

million biosignal images spanning several modalities and tasks [35]. Representation design has also been studied directly in the visual domain, including synthetic thermal and RGB videos for pain assessment [36] and reorganized facial spatiotemporal representations intended to preserve more useful structure for recognition [37]. These studies either spatially composite representations, stack signal images outside the EDA stress-detection setting, or learn a common embedding space across modality-specific images. The same question has also been explored in the temporal domain. Multi-window respiration models have combined short and long signal segments for pain recognition [38], while multi-scale EEG models have used sample-adaptive fusion across several window lengths for emotion recognition [39]. A systematic comparison of the raw EDA waveform, multiple EDA-derived visual representations, and channel-stacked combinations under a common subject-level stress-recognition protocol remains missing.

III. METHODOLOGY

A. EDA Signal Representations

Each EDA recording is a single-channel signal sampled at 1 kHz throughout the full 120-second task duration, yielding 119,988 raw samples. Seven representations are derived from the raw signal. The first is the raw waveform itself, used both as a standalone 1D input and as one channel in every combined configuration. The other six are 2D image representations computed after downsampling the signal to $f_{ds} = 50$ Hz, matching the approximate 0.05–5 Hz bandwidth of EDA. The six visual representations are defined as follows. (1) *PSD*. A magnitude spectrogram is computed via short-time Fourier transform (STFT) with a 128-sample window and 120-sample overlap, band-limited to 0.05–5 Hz and log-compressed. (2) *Angle*. The unwrapped phase of the same STFT is computed by applying phase unwrapping along the time axis, removing the wrapping discontinuities that would otherwise appear as a checkerboard artifact on the slowly varying EDA signal. (3) *Phase*. An instantaneous-frequency map is derived from the time derivative of the unwrapped phase, with time-frequency bins below the 50th percentile magnitude masked to zero and the result smoothed with a Gaussian filter ($\sigma = 1.5$) to suppress noise in low-power regions. (4) *Scalogram*. A continuous wavelet transform is applied using a Morlet wavelet across 128 log-spaced scales corresponding to the 0.05–5 Hz band, producing a multi-resolution time-frequency energy map. (5) *Recurrence*. A recurrence plot is computed from the z-score normalized signal using a point threshold at the 20th percentile, visualizing recurring states in the signal’s dynamics. (6) *Wave*. The full-resolution waveform is rendered directly as a line plot, converting the 1D trace into a 2D image without any time-frequency or nonlinear-dynamics transformation. All six image representations are rendered at 224×224 resolution using the same colormap and canvas-capture pipeline, with axes stripped before saving as lossless images. *PSD*, *Phase*, and *Scalogram* are additionally contrast-normalized to the 5th–95th percentile range, while *Angle* is rendered without percentile clipping. The shared rendering pipeline mitigates differences

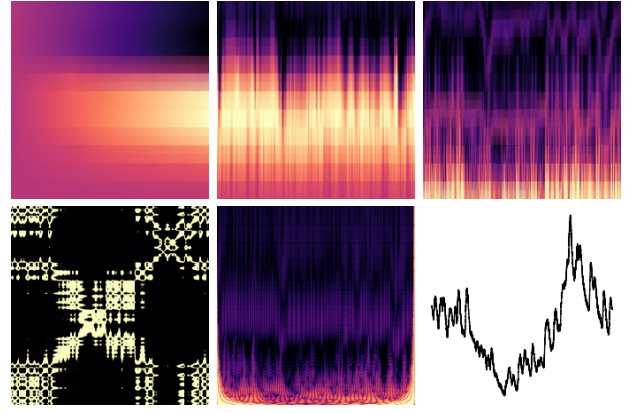


Fig. 1: Example of the six visual representations derived from a single EDA recording. Top row (left to right): *Angle*, *Phase*, *PSD*. Bottom row (left to right): *Recurrence*, *Scalogram*, *Wave*.

due to image style and keeps the comparison focused on the underlying signal transformation. Figure 1 shows one example of each representation computed from the same recording.

B. Multi-Representation Fusion

Representations are combined by channel stacking rather than by spatial concatenation or separate encoder branches. Each selected image representation is flattened from its native $224 \times 224 \times 3$ shape into a one-dimensional vector of length 150,528. It is then stacked as an additional channel alongside the raw waveform, which is zero-padded from its native length of 119,988 samples to the common length $L = 150,528$ whenever at least one image representation is included. A configuration with R image representations produces a tensor $\mathbf{X} \in \mathbb{R}^{B \times C \times L}$ with $C = 1 + R$ channels. This tensor is processed as a 1D input by the same architecture used for the raw waveform alone, analogous to a multichannel physiological signal such as EEG, where recording channels are stacked along C . Channel-wise layer normalization is applied before the backbone, so that raw EDA amplitudes and flattened pixel-intensity channels are normalized independently instead of being forced to share a common statistic. When a single image representation is evaluated without the waveform, the native $224 \times 224 \times 3$ image is processed directly as a 2D input, following Section III-C. In the rest of the paper, waveform $\oplus$ *Angle* denotes the channel-stacked combination of the two representations. The symbol $\oplus$ is used to distinguish channel stacking from elementwise addition, which is not performed here.

C. Asymmetric Attention

For either input type, geometric information is added by encoding each token position with Fourier features using $K = 6$ frequency bands and a maximum frequency $f_{\max} = 10$. The token sequence, whether the $N = H \times W = 50,176$ tokens of a single 224×224 image or the L positions of the raw waveform or a channel-stacked combination, is partitioned into $S = 32$ contiguous segments. These segments divide the input

along the token axis and do not correspond to external temporal or spatial windows of the original signal.

The same single-layer backbone is used for every configuration: the raw waveform, each individual image-based representation, and every channel-stacked combination. It consists of one cross-attention block followed by $R = 8$ self-attention blocks. A single latent state is associated with each segment and is instantiated at runtime by replicating a shared initialization vector derived from a set of $M_0 = 64$ learnable global parameters $\{\ell_m\}_{m=1}^{M_0}$:

$$\ell_{\text{init}} = \frac{1}{M_0} \sum_{m=1}^{M_0} \ell_m \in \mathbb{R}^{d_0}, \quad (1)$$

where $d_0 = 128$. These segment states are not independently learnable; segment-specific representations emerge through the attention updates.

1) *Cross-attention.*: Each segment state aggregates information exclusively from its corresponding token subset through cross-attention:

$$\mathbf{e}_s = \mathbf{e}_s^{(0)} + \text{Attn}(\mathbf{e}_s^{(0)}, \tilde{\mathbf{T}}_s), \quad (2)$$

where $\mathbf{e}_s^{(0)} \in \mathbb{R}^{B \times 1 \times d_0}$ provides the queries and $\tilde{\mathbf{T}}_s \in \mathbb{R}^{B \times n_s \times C'}$ provides the keys and values. This operation is asymmetric: the query side consists of a single vector of dimension d_0 , while the key-value side spans $n_s \gg 1$ token vectors of dimension C' . The resulting attention matrix is $1 \times n_s$, not square, and the query and key-value spaces differ in both size and dimensionality. All $S = 32$ segments are processed in parallel by packing into the batch dimension, without altering the underlying computation. Cross-attention uses 8 heads, each of dimension 16.

2) *Self-attention.*: After cross-attention, all segment states are stacked to form the segment-state matrix $\mathbf{E} \in \mathbb{R}^{B \times S \times d_0}$. Self-attention is applied across all S segment states, enabling global information exchange:

$$\mathbf{E} \leftarrow \mathbf{E} + \text{Attn}(\mathbf{E}, \mathbf{E}), \quad (3)$$

repeated $R = 8$ times. Self-attention uses 8 heads, each of dimension 16.

After the self-attention blocks, the final segment states $\mathbf{E} \in \mathbb{R}^{B \times S \times 128}$ are averaged across segments and passed through a linear classification head. Both attention operations use pre-layer normalization and residual connections, with attention and feedforward dropout of 0.10 applied uniformly. The complete set of architectural hyperparameters, shared by every configuration evaluated in this study, is reported in Table I, and the full pipeline is illustrated in Fig. 2.

D. Augmentation, Regularization, and Training

The waveform and image representations have different tensor shapes, so each uses an augmentation pipeline matched to its native format. In combined configurations, the waveform channel is augmented with the 1D pipeline, while each image channel is augmented with the 2D pipeline before flattening and channel stacking. This preserves the natural domain of

TABLE I: Architectural hyperparameters, shared by every configuration evaluated in this study.

Hyperparameter	Value
Depth	1
Number of latents	64
Latent dimension (d_0)	128
Cross-attention heads	8
Cross-attention head dimension	16
Self-attention heads	8
Self-attention head dimension	16
Self-attention blocks per cross (R)	8
Segments (S)	32
Attention dropout	0.10
Feedforward dropout	0.10
Fourier frequency bands (K)	6
Maximum frequency (f_{max})	10

This single-layer backbone is identical across every representation and combination; differences in Params (M) and GFLOPs across rows in Table V come entirely from differences in input shape (1D waveform length vs. 2D image size vs. number of stacked channels), not from any change in architecture.

each transformation during augmentation. Label smoothing, attention dropout, feed-forward dropout, optimizer, learning-rate schedule, and batch size are shared across all configurations. The complete setup is reported in Table II.

IV. EXPERIMENTAL EVALUATION & RESULTS

This section evaluates the choice of representation and fusion for EDA-based stress detection. All experiments use binary classification. Validation performance is reported with macro-averaged accuracy, precision, and F1 score, together with their arithmetic mean, denoted *Average*. Test performance is reported with macro-averaged accuracy and is used as the primary comparison criterion.

A. Dataset and Protocol

This study uses a stress dataset of 58 adults (24 men, 34 women), aged 26.9 ± 4.8 years. The protocol includes four stress-induction phases: social exposure, emotional recall, mental workload, and stressful video stimuli. Each participant completed 11 tasks: 4 neutral, 6 stress-inducing, and 1 relaxation task, as reported in Table III. Social exposure consisted of a psychologist-led interview focused on negative personality traits. Emotional recall required participants to relive a past stressful event in real time. Mental workload was induced with a modified Stroop Color-Word Test [40] and the Paced Auditory Serial Addition Test [41]. The stressful stimuli phase used videos depicting accidents and acrophobia; a relaxing video was included as a physiological recovery baseline between induction phases and was excluded from binary classification. Stress induction was verified through Heart Rate monitoring, which showed a statistically significant increase during stress tasks ($p < 0.05$). Subjective validation was performed using Self-Assessment Manikin scales, in which participants reported higher arousal and lower valence during stressful phases than during neutral baselines. Electrodermal activity was recorded continuously from a single channel at

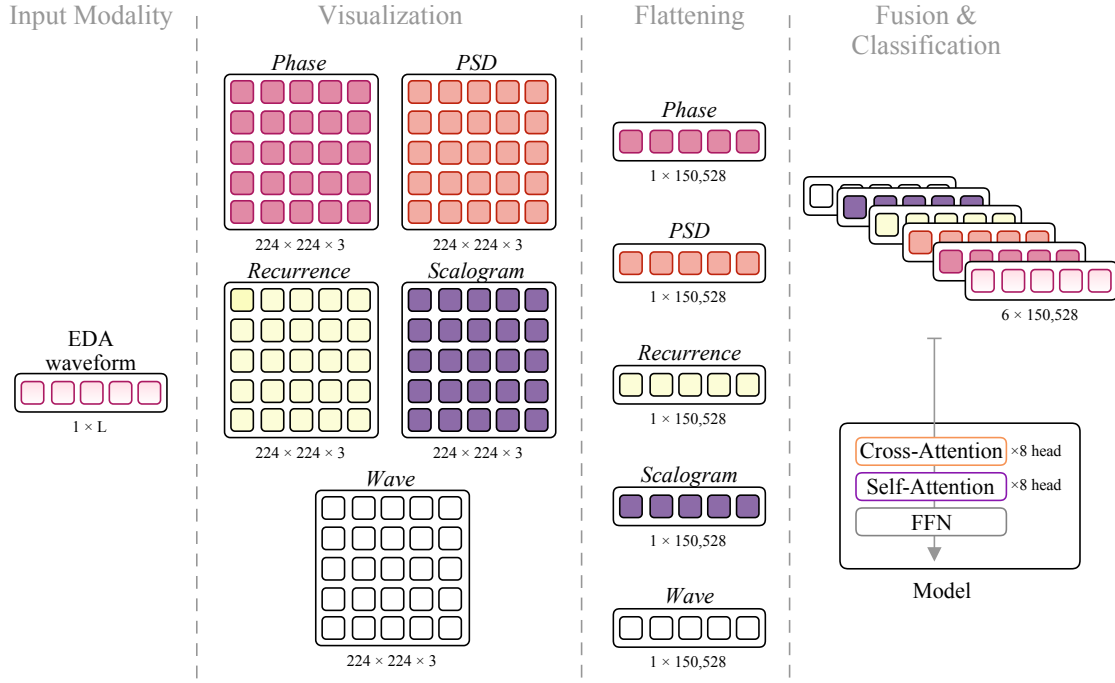


Fig. 2: Overview of the proposed pipeline. Each EDA recording produces the raw waveform and, in parallel, six visual representations. Representations are flattened into one-dimensional channels, stacked together with the raw waveform, and processed by the shared asymmetric-attention backbone for binary stress classification.

1 kHz and was the only input modality used here. The task is binary classification between neutral and stress conditions. The study received approval from the local Research Ethics Committee (approval no. 155/12-09-2022), and all participants provided informed consent. The dataset is available for non-commercial research upon request.¹ Subjects are split at the subject level, with no participant appearing in more than one set. To reduce performance inflation due to subject-difficulty imbalance, a stratified split protocol is used: subjects are ranked by the combined z-score from leave-one-subject-out difficulty estimation and assigned to four quartiles. The final partition contains 38 training, 8 validation, and 12 testing subjects, with all sets containing subjects from all four difficulty groups in proportion. Table IV reports the exact split. The resulting data contain 414 training samples (188 neutral, 226 stress), 83 validation samples (37 neutral, 46 stress), and 132 testing samples (60 neutral, 72 stress).

B. Representation Comparison

Table V reports performance and computational cost for every single representation and every combined configuration after 400 training epochs. The raw waveform reaches 67.36% test accuracy. Seven of the eight combined configurations exceed this value; the only exception is waveform $\oplus$ Phase, which drops to 64.72%. The best configuration combines the waveform with PSD, Recurrence, Scalogram, and Wave, while excluding Angle. It reaches 70.97%, improving the raw

waveform by 3.61 points. Fusing all six representations also improves over the raw waveform, reaching 68.89%, but remains below the excluded-Angle configuration.

Among individual representations, PSD is the strongest on test accuracy, reaching 69.44%. It remains within 1.53 points of the best fused configuration while requiring only one representation. Wave and waveform $\oplus$ Wave both reach 68.89%, showing that a rendered image of the waveform trace is competitive even though it contains no new signal source. Angle is the weakest single representation, with 57.78% test accuracy, roughly 10 points below the other single inputs. This result motivates its exclusion from the best-performing fusion.

All results above use the 400-epoch training budget. Table VI reports test accuracy at 200 and 400 epochs for each configuration. The longer budget was introduced after preliminary experiments showed that some representations, especially Wave, continued to improve after 200 epochs.

C. Overall Analysis & Discussion

Channel-stacked fusion gives the best test accuracy in this study, but the margin over the strongest single representation is modest. The excluded-Angle configuration achieves 70.97% on the test set, while PSD alone achieves 69.44%, a difference of 1.53 points at roughly half the GFLOPs. Fusion of all six representations reaches 68.89%, the same as Wave alone. Fusion is therefore not automatically the best practical choice. In accuracy-critical settings, the excluded-Angle configuration is preferred; in compute-constrained settings, a carefully selected single representation remains competitive.

¹https://github.com/ggian/stress_dataset

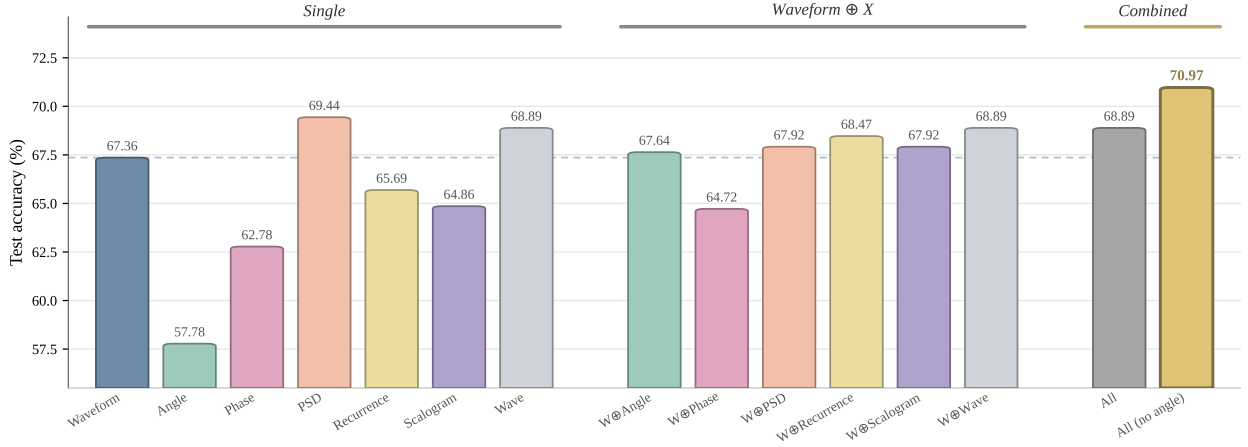


Fig. 3: Test accuracy across single representations, waveform $\oplus$ representation combinations, and the two full-fusion configurations, at the 400-epoch training budget. The excluded-*Angle* configuration reaches the highest test accuracy at 70.97%; *PSD* alone remains within 1.53 points at 0.86 vs. 1.58 GFLOPs.

TABLE II: Augmentation, regularization, and training configuration.

Method / Parameter	Value
<i>Waveform (1D) augmentations</i>	
AddNoise	$p \in [0.10, 0.50]$, SNR factor $\in [1, 1000]$
Masking	$p \in [0.10, 0.50]$, mask 15–30% of length
<i>Image (2D) augmentations</i>	
AugMix	$p = 0.10$, magnitude 3
TrivialAugmentWide	$p = 0.10$, 31 magnitude bins
CenterCrop	$p = 0.10$, crop 200×200 , resize 224×224
Noise	$p = 0.10$, $\sigma = 100$
Masking	$p = 0.10$, 20 blocks of 32×32
<i>Regularization</i>	
Label Smoothing	0.10
Att-Dropout	0.10
FF-Dropout	0.10
<i>Training</i>	
Optimizer	AdamW
Learning rate	$1e-4$
LR schedule	cosine
Weight decay	0.05
Warmup epochs	20
Cooldown epochs	10
Batch size	32
Epochs	200/400

Waveform augmentations are applied to the 119,988-sample raw signal; image augmentations are applied to the 224×224 PIL image. Both are applied during training only; validation and testing use no augmentation. In combined configurations, each channel type is augmented in its native domain before flattening and channel stacking. Att-Dropout/FF-Dropout: dropout probability in attention/feed-forward sublayers. $p \in [x_1, x_2]$ denotes $p \sim \mathcal{U}(x_1, x_2)$ sampled per sample.

TABLE III: Experimental tasks employed in this study.

#	Task	Duration (sec)	State
<i>Social Exposure</i>			
1	Neutral reference	120	N
2	Baseline description	120	N
3	Interview	120	S
<i>Emotional Recall</i>			
4	Neutral reference	120	N
5	Recall stressful event	120	S
<i>Mental Workload</i>			
6	Reading reference	120	N
7	Stroop Color-Word Test	120	S
8	PASAT task	120	S
<i>Stressful Stimuli</i>			
9	Relaxing video*	120	R
10	Adventure video	120	S
11	Psychological pressure	120	S

N=neutral S=stress R=relaxed. *: Used as a physiological recovery baseline between induction phases; excluded from binary classification.

We note that in this study, since all configurations were evaluated using the same fixed subject-independent split, the results are treated as a benchmark comparison, with held-out test accuracy as the primary metric. Under this protocol, the excluded-*Angle* fusion achieved the highest test accuracy. Its 1.53-point improvement over *PSD*, however, should be interpreted only within this experimental setting and not as evidence that the same configuration will necessarily remain superior across other datasets, splits, or architectures.

The results also show that fusion is not uniformly beneficial. Waveform $\oplus$ *Phase* reaches 64.72%, below the raw waveform at 67.36%, and waveform $\oplus$ *Angle* improves the waveform only marginally, from 67.36% to 67.64%. The weaker fused configurations are those built around *Phase* and *Angle*, the two STFT-phase-derived representations. In this setting, phase-based encodings do not add useful complementary information

TABLE IV: Subject-level split by difficulty group. Subjects are ranked by combined z-score and assigned to four quartile-based groups (Q1 = hardest, Q4 = easiest).

Split	Difficulty Group			
	Q1 – Hard	Q2 – Med-Hard	Q3 – Med-Easy	Q4 – Easy
Training (38)	P017, P018, P022, P026, P034, P035, P042, P045, P050, P056	P001, P002, P003, P007, P012, P021, P033, P040, P048	P004, P014, P016, P032, P036, P046, P047, P052, P053, P054	P005, P010, P020, P028, P029, P037, P039, P041, P057
Validation (8)	P038, P055	P009, P023	P006, P013	P019, P030
Testing (12)	P008, P025, P044	P011, P024, P043	P015, P031, P058	P027, P051, P059

Q1: $z < -0.46$; Q2: $-0.46 \leq z < -0.05$; Q3: $-0.05 \leq z < +0.40$; Q4: $z \geq +0.40$.

TABLE V: Performance and computational cost using the EDA modality (400 epochs).

Representation	Computational Cost		Validation				Testing
	Params (M)	GFLOPs	Accuracy	Precision	F1	<i>Average</i>	Accuracy
Waveform	2.00	0.97	61.78	61.66	61.40	<i>61.61</i>	67.36
Angle	1.86	0.86	66.13	66.00	66.02	<i>66.05</i>	57.78
Phase	1.86	0.86	71.30	72.21	71.48	<i>71.66</i>	62.78
PSD	1.86	0.86	68.33	70.26	68.39	<i>68.99</i>	69.44
Recurrence	1.86	0.86	66.42	67.13	66.52	<i>66.69</i>	65.69
Scalogram	1.86	0.86	65.63	67.95	65.46	<i>66.35</i>	64.86
Wave	1.86	0.86	66.42	67.13	66.52	<i>66.69</i>	68.89
Waveform $\oplus$ Angle	2.06	1.27	62.31	62.38	61.44	<i>62.04</i>	67.64
Waveform $\oplus$ Phase	2.06	1.27	63.40	63.40	62.65	<i>63.15</i>	64.72
Waveform $\oplus$ PSD	2.06	1.27	66.42	67.13	66.52	<i>66.69</i>	67.92
Waveform $\oplus$ Recurrence	2.06	1.27	60.75	62.35	60.35	<i>61.15</i>	68.47
Waveform $\oplus$ Scalogram	2.06	1.27	67.51	68.49	67.62	<i>67.87</i>	67.92
Waveform $\oplus$ Wave	2.06	1.27	66.42	67.13	66.52	<i>66.69</i>	68.89
All	2.06	1.66	68.60	69.91	68.72	<i>69.08</i>	68.89
All (no angle)	2.06	1.58	65.31	65.13	64.98	<i>65.14</i>	70.97

Average: arithmetic mean of validation Accuracy, Precision, and F1. The bold value indicates the highest overall test accuracy. Params (M) and GFLOPs for the visual representations are identical to *Angle*: all six are $224 \times 224 \times 3$ images processed by the identical backbone (Table I). *Waveform* uses the same backbone throughout; its different Params/GFLOPs reflect its different input shape (a 1D sequence rather than a 2D image), not a different architecture.

TABLE VI: Test accuracy at 200 vs. 400 training epochs.

Representation	200 ep.	400 ep.	Δ
Waveform	67.08	67.36	+0.28
Angle	57.78	57.78	+0.00
Phase	61.53	62.78	+1.25
PSD	69.44	69.44	+0.00
Recurrence	63.75	65.69	+1.94
Scalogram	63.75	64.86	+1.11
Wave	58.89	68.89	+10.00
Waveform $\oplus$ Angle	69.03	67.64	−1.39
Waveform $\oplus$ Phase	63.47	64.72	+1.25
Waveform $\oplus$ PSD	66.67	67.92	+1.25
Waveform $\oplus$ Recurrence	66.81	68.47	+1.66
Waveform $\oplus$ Scalogram	64.31	67.92	+3.61
Waveform $\oplus$ Wave	68.75	68.89	+0.14
All	68.19	68.89	+0.70
All (no angle)	69.44	70.97	+1.53

when stacked with the raw waveform and can degrade the fused input.

The number of parameters and GFLOPs reported in Table V

include the classification model only. The visual representations are generated offline using standard signal-processing operations that can run on a conventional CPU and do not require GPU acceleration. They nevertheless introduce additional preprocessing and storage overhead. Since these costs were not measured, the reported efficiency comparison is limited to model complexity and inference computation.

The exclusion of *Angle* from the strongest configuration follows from its empirical behavior. *Angle* has the lowest test accuracy among all single representations, roughly 10 points below the next weakest single input. It is also present in the two combined configurations, waveform $\oplus$ *Angle* and All, that remain below *All (no angle)*. Removing it therefore addresses a specific failure of this representation rather than imposing a general rule about which signal views should be included.

The *Wave* representation reaches 68.89% on its own and the same value when fused with the waveform. This representation contains the same underlying information as the raw waveform, but in a rendered 2D form. Its performance indicates that the choice of representation affects the inductive bias available to the model, even when the original signal content remains unchanged. In other words, the way the signal is presented to

the architecture matters.

Several limitations should be noted. First, the six representations were evaluated with fixed transformation parameters. Tuning the STFT window, wavelet scales, recurrence threshold, or rendering parameters could improve specific representations and alter the observed ranking. Second, the channel order in the stacked input was fixed. The attention-based backbone can, in principle, route information across channels regardless of their positions, but this was not tested directly. Third, the study uses a single dataset and a binary neutral-versus-stress classification setting. The ranking of representations, the benefit of fusion, and the exclusion of *Angle* should therefore be tested on additional stress and affective-computing datasets.

V. CONCLUSION

This paper examined representation choice and fusion for EDA-based stress detection. The raw one-dimensional waveform and six image-based signal representations were evaluated individually and in channel-stacked combinations. All configurations used the same asymmetric-attention backbone, isolating the effect of the input representation from changes in architecture. Representation fusion improved over the raw waveform in seven of eight combined configurations. The best result was obtained with the excluded-*Angle* configuration, achieving 70.97% test accuracy, a 3.61-point improvement over the raw waveform at 67.36%. The *PSD* spectrogram achieved 69.44% as a single representation, remaining close to the best fused configuration while incurring lower computational cost. The unwrapped phase spectrogram, *Angle*, was the only representation that consistently weakened fusion and was excluded from the best-performing combination based on its test behavior. Future work should examine per-representation parameter tuning, longer or representation-specific training schedules, alternative channel-ordering strategies, and evaluation on additional stress datasets. The results provide a reference point for studying biosignal representations in stress detection and show that the choice of representation can be as important as the model architecture itself.

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